KAIZEN MATHS

Lesson starter

Loci and Constructions

Regions

Name _____

SA1

Construct the two boundaries, then identify the region satisfying both conditions.



A point P is less than 3 cm from A and closer to A than to B. Describe the region where P can be.

GEO-F01A-L3 · SA

Show your working

KAIZEN MATHS

Lesson starter

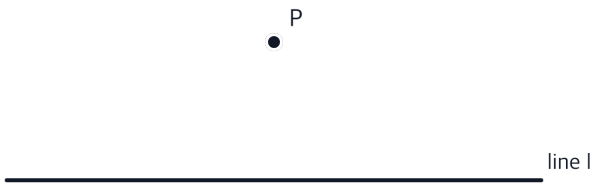
Loci and Constructions

Regions

Name _____

SB1

Construct the two boundaries, then identify the region satisfying both conditions.



A point X is within 3 cm of line l and more than 4 cm from point P. Describe the region where X can be.

GEO-F01A-L3 · SB

Show your working

KAIZEN MATHS

Lesson starter

Loci and Constructions

Regions

Name _____

SC1

Construct both distance circles, then identify the possible intersection positions.



Point X is 5 cm from A and 5 cm from B. Show the possible positions of X.

GEO-F01A-L3 · SC

Show your working

KAIZEN MATHS

Lesson starter

Loci and Constructions

Regions

Name _____

SD1

Construct the two boundaries, then identify the region satisfying both conditions.



A point P is less than 5 cm from A and closer to A than to B. Describe the region where P can be.

GEO-F01A-L3 · SD

Show your working

KAIZEN MATHS

Exit ticket

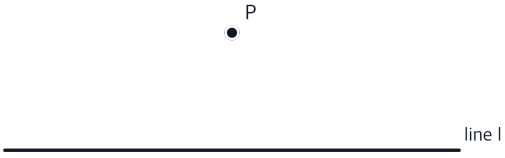
Loci and Constructions

Regions

Name _____

EA1

Construct the two boundaries, then identify the region satisfying both conditions.



A point X is within 4 cm of line l and more than 5 cm from point P. Describe the region where X can be.

Show your working

KAIZEN MATHS

Exit ticket

Loci and Constructions

Regions

Name _____

EB1

Construct both distance circles, then identify the possible intersection positions.



Point X is 7 cm from A and 6 cm from B. Show the possible positions of X.

Show your working

KAIZEN MATHS

Exit ticket

Loci and Constructions

Regions

Name _____

EC1

Construct the two boundaries, then identify the region satisfying both conditions.



A point P is less than 4 cm from A and closer to A than to B. Describe the region where P can be.

Show your working

KAIZEN MATHS

Exit ticket

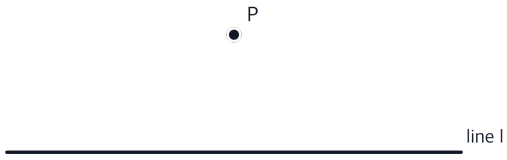
Loci and Constructions

Regions

Name _____

EA1

Construct the two boundaries, then identify the region satisfying both conditions.



A point X is within 4 cm of line l and more than 5 cm from point P. Describe the region where X can be.

Show your working

Teacher answers & working

Loci and Constructions · Regions

GEO-F01A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Construct the two boundaries, then identify the region satisfying both conditions.



A point P is less than 3 cm from A and closer to A than to B. Describe the region where P can be.

Answer:

Inside the circle centre A, radius 3 cm, on the A side of the perpendicular bisector of AB.

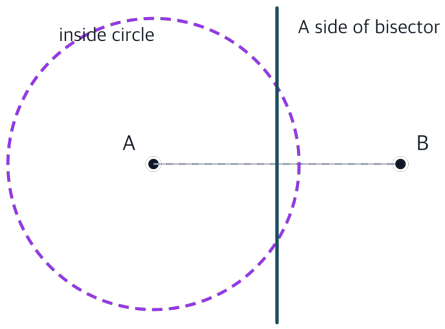
Working:

Less than 3 cm from A means inside the circle with centre A and radius 3 cm.

Closer to A than to B is decided by the perpendicular bisector of AB.

Construct the perpendicular bisector of AB.

Choose the side of the perpendicular bisector that contains A.



The required region is inside the circle centre A, radius 3 cm, and on the A side of the perpendicular bisector.

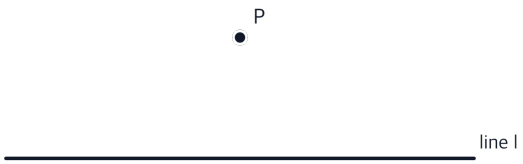
Teacher answers & working

Loci and Constructions · Regions

GEO-F01A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

Construct the two boundaries, then identify the region satisfying both conditions.



A point X is within 3 cm of line l and more than 4 cm from point P. Describe the region where X can be.

Answer:

Between the two parallel lines 3 cm from l, but outside the circle centre P, radius 4 cm.

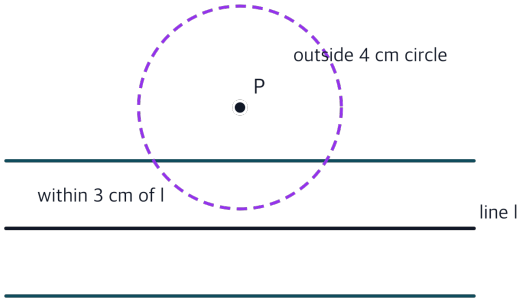
Working:

Within 3 cm of line l means between two parallel lines, each 3 cm from l.

More than 4 cm from P means outside the circle with centre P and radius 4 cm.

Draw both parallel boundary lines.

Draw the circle boundary around P.



The required region is inside the parallel-line band and outside the circle.

Teacher answers & working

Loci and Constructions · Regions

GEO-F01A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

Construct both distance circles, then identify the possible intersection positions.



Point X is 5 cm from A and 5 cm from B. Show the possible positions of X.

Answer:

The possible positions are the intersection points of the two circles.

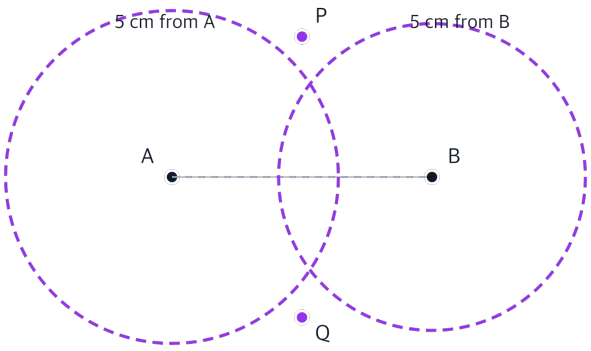
Working:

Draw a circle with centre A and radius 5 cm.

Draw a circle with centre B and radius 5 cm.

A point satisfying both conditions must lie on both circles.

Therefore the possible positions are where the two circles intersect.



There are two possible positions unless the circles touch once or do not meet.

Teacher answers & working

Loci and Constructions · Regions

GEO-F01A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1

Construct the two boundaries, then identify the region satisfying both conditions.



A point P is less than 5 cm from A and closer to A than to B. Describe the region where P can be.

Answer:

Inside the circle centre A, radius 5 cm, on the A side of the perpendicular bisector of AB.

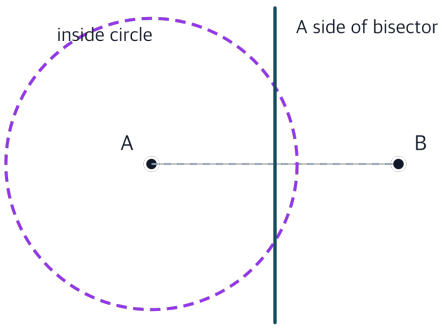
Working:

Less than 5 cm from A means inside the circle with centre A and radius 5 cm.

Closer to A than to B is decided by the perpendicular bisector of AB.

Construct the perpendicular bisector of AB.

Choose the side of the perpendicular bisector that contains A.



The required region is inside the circle centre A, radius 5 cm, and on the A side of the perpendicular bisector.

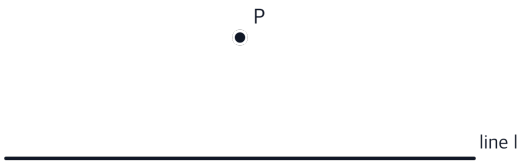
Teacher answers & working

Loci and Constructions · Regions

GEO-F01A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Construct the two boundaries, then identify the region satisfying both conditions.



A point X is within 4 cm of line l and more than 5 cm from point P. Describe the region where X can be.

Answer:

Between the two parallel lines 4 cm from l, but outside the circle centre P, radius 5 cm.

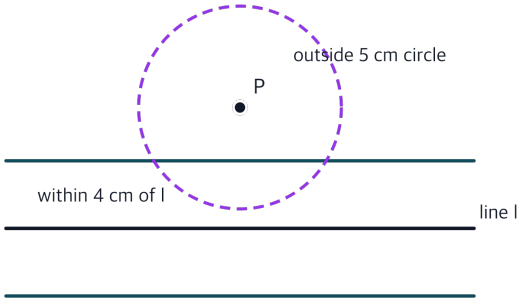
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Draw both parallel boundary lines.

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Teacher answers & working

Loci and Constructions · Regions

GEO-F01A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

Construct both distance circles, then identify the possible intersection positions.



Point X is 7 cm from A and 6 cm from B. Show the possible positions of X.

Answer:

The possible positions are the intersection points of the two circles.

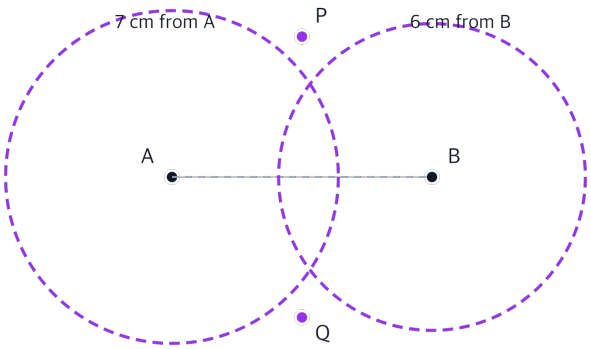
Working:

Draw a circle with centre A and radius 7 cm.

Draw a circle with centre B and radius 6 cm.

A point satisfying both conditions must lie on both circles.

Therefore the possible positions are where the two circles intersect.



There are two possible positions unless the circles touch once or do not meet.

Teacher answers & working

Loci and Constructions · Regions

GEO-F01A-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1

Construct the two boundaries, then identify the region satisfying both conditions.



A point P is less than 4 cm from A and closer to A than to B. Describe the region where P can be.

Answer:

Inside the circle centre A, radius 4 cm, on the A side of the perpendicular bisector of AB.

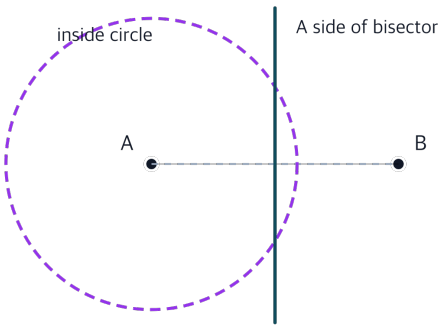
Working:

Less than 4 cm from A means inside the circle with centre A and radius 4 cm.

Closer to A than to B is decided by the perpendicular bisector of AB.

Construct the perpendicular bisector of AB.

Choose the side of the perpendicular bisector that contains A.



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